

IDENTIFICATION

CODE : GI-4-S1-EC-SIE
ECTS : 3

HOURS

Cours : 6h
TD : 16h
TP : 18h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 42h
Travail personnel : 18h
Total : 62h

ASSESSMENT METHOD

Situational assessment Group score

TEACHING AIDS

Project Subject, Company Data
Readings to be done before some sessions (theoretical complements)
Methodological guides presented during the sessions

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to the Management and Exploitation of Company Data (GI-4-S1-UE-GEDE) teaching unit and contributes to the following skills:

By allowing the student to work and be evaluated on the following knowledge:

- BPMN modeling
- Process Value Analysis
- Evaluation of the performance of a process
- Process mining
- The various industrial information systems

By allowing the student to work and be evaluated on the following abilities:

- Model a complex business process
- Diagnose a business process in terms of inefficiencies and non-value-added tasks/flows
- Identify and evaluate business process improvement solutions
- Specify the information system supporting a business process

CONTENT

This course is conducted as a team project and is based on an industrial case. The objective of the project is to carry out a preliminary study for the modernization of a manufacturing company's information system. This preliminary study includes several phases: (1) Analysis of the existing situation (AS-IS) through the modeling of current business processes, the diagnosis of these processes and the identification of improvement solutions; (2) Design of an improvement solution (TO-BE) through the modeling of new processes and the specification of a software/hardware solution to support the new organization (without however developing it).

BIBLIOGRAPHY

1. Marlon Dumas, Marcello La Rosa, Jan Mendling, Hajo A. Reijers. "Fundamentals of Business Process Management". Editions Springer, ISBN :9783662565094 / 3662565099, 2018, 527 pages
2. Heru Susanto, Fang-Yie Leu, Chin Kang Chen. "Business Process Reengineering: An ICT Approach". Editions Apple Academic Press, ISBN :9780429949302 / 0429949308, 2019, 248 pages
3. W.M.P. van der Aalst. "Process Mining: Data Science in Action". Editions Springer-Verlag, 2016. <http://www.springer.com/978-3-662-49850-7>

PRE-REQUISITES

- Process Modeling (BPMN)
- Object-oriented modeling (UML)